

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MRBENCH | Context: Metropolitan India — Grades 1–8 Mathematics Tutoring (Student Population)

Definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Justification

Input content is a HIGH-priority dimension per elicitation, and the deployment requires Indian everyday contexts (cricket, rupees, local markets, school fees) with explicit rejection of Western contexts (A2). MRBench's 192 instances are drawn entirely from Bridge and MathDial [Q41, Q52], neither of which is documented as containing Indian curriculum content, Indian numeral conventions, or India-specific scenarios. The paper notes that Bridge problems are 'fundamental mathematical concepts, including operations such as multiplication, addition' [Q45] in short Western-style framings, and MathDial is 'middle school-level mathematical reasoning' [Q49] without regional grounding. Web search confirms no Indian-context tutoring dialogue dataset exists [WEB-6, WEB-7], and GSM8K robustness research (Tomar et al. 2025) directly demonstrates measurable LLM performance drops when Western contexts are replaced with Indian ones [WEB-8] — establishing that this is not cosmetic but a measurable validity threat. There is no evidence of any India-relevant content in MRBench's instance pool.

ID	Evidence
WEB-6	No NCERT-aligned Grade 1–8 English-medium tutoring dialogue benchmark exists
WEB-7	HAWP is Hindi arithmetic word problems only, not tutoring dialogue
WEB-8	GSM8K robustness research shows Indian-context substitution causes measurable LLM performance drops

Strengths

- Instance pool is rigorously filtered for high quality, with a clear criterion that the student's last utterance contains a mistake [Q46, Q47, Q51]
- Combines elementary and middle-school-level dialogues providing some grade-banding diversity within math [Q97, Q143, Q144]
- Fundamental mathematical operations covered (addition, multiplication) overlap with NCERT primary content even if framing differs [Q45]

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires Indian cultural, geographic, and curricular grounding (NCERT topics, rupees, cricket, local markets, train timetables, school fees, festival contexts). Confirmed via elicitation A1, A2 and NCERT syllabus sources [WEB-2, WEB-4].

ID	Evidence
WEB-2	NCERT Grade 1–5 mathematics syllabus (place value with lakhs, spiral sequencing)

WEB-4	NCERT upper-primary syllabus emphasizing abstraction, fractions, integers, algebra
-------	--

IC-2: No — there is no evidence that MRBench's source datasets contain any India-aligned cultural content. Bridge and MathDial are sourced from Western/general-context tutoring corpora [Q41, Q44, Q48] with no documentation of Indian inclusion.

IC-3: Likely yes. The paper notes Bridge dialogues focus on 'fundamental operations' [Q45] in short Western-framed contexts, and MathDial's middle-school reasoning problems [Q49] are not documented as having Indian grounding. GSM8K robustness research directly demonstrates that Western-to-Indian context substitution degrades LLM performance [WEB-8], so Western framings in MRBench's instances do not transfer cleanly.

ID	Evidence
WEB-8	GSM8K robustness research shows Indian-context substitution causes measurable LLM performance drops

IC-4: Not performed by the benchmark authors. INSUFFICIENT DOCUMENTATION on whether any regional annotators reviewed instance content for cultural fit; the four annotators' nationalities are undisclosed [Q31].

IC-5: Documented harms to content validity: (a) no rupee-denominated problems; (b) no Indian numeral conventions in problem statements; (c) no Indian everyday scenarios; (d) Western framings that empirically degrade LLM performance when transposed [WEB-8]. The deployment's HIGH-priority requirement for Indian-context word problems is entirely unmet.

ID	Evidence
WEB-6	No NCERT-aligned Grade 1–8 English-medium tutoring dialogue benchmark exists
WEB-8	GSM8K robustness research shows Indian-context substitution causes measurable LLM performance drops

Information Gaps

- No instance-level content audit of Bridge/MathDial for cultural framings is published
- No data on whether any of the 192 instances incidentally contain India-applicable framings

Requires Expert Verification

- Sample-level review by Indian educators to confirm degree of cultural distance in actual instance content
- Empirical measurement of LLM performance drop on MRBench instances when transposed to Indian context

Remediation

Gap 1: All 192 instances are drawn from Western-context Bridge and MathDial datasets with no documented Indian content; GSM8K research shows context substitution causes measurable LLM performance drops.

Recommendation: Construct a parallel evaluation set of NCERT-aligned dialogues with Indian everyday contexts (cricket, rupees, local markets, school fees, train timetables) authored or reviewed by Indian primary/middle-school teachers, and re-evaluate the seven LLMs on this set to quantify the regional performance gap.

Gap 2: DPDP Act 2023 / Rules 2025 impose verifiable parental consent and ban profiling/tracking of under-18 users [WEB-23, WEB-24], constraining how Indian student data can be collected for any supplementation effort.

Recommendation: Design any Indian content-collection or re-annotation effort under DPDP-compliant verifiable parental consent workflows; avoid behavioural-monitoring data collection for under-18 users and document data fiduciary obligations before deployment.